The cyclotomic γ-weight base's falsifier

Encyclopedia §8 entry D4, lead A2 — 2026-10-15

`research/cyclotomic_falsifier_20261015`

Verdict (pre-registered rule, PROTOCOL.md frozen at `9972ea0e8`). The **yield clause** of A2 ("fail if its yield per column is lower") **survives, yield-neutral**: on all 15 $n = 23$ tests the cyclotomic base C_w is inside the distribution of 1999 random Frobenius-stable bases with the same columns (effects −0.49% … +0.65%, smallest p = 0.363, critical 0.00333), while a planted positive control is detected on every cell (p = 0.001; −35% … −64% coverage, +246% … +293% energy). D4 therefore **survives** only in the sense that its yield is not lower; it is not higher either. Any value of the lead rests on the untested **solver clause**, whose only record ($n = 11$) points the wrong way. Cost (ecbench, operations only, L0): cyclotomic vs compact-orbit-scan at equal columns is costlier in 2 of 4 pairs (+2.2%, +12.2%, factor-base build phase) and unresolved in 2.

1. Question

A2 (`docs/ic/RESEARCH_IC_NOVEL_DIRECTIONS_20261007.md` L187–221): $C_w = \left\{\sum_{k\in S} \gamma^k : |S| = w\right\}$, γ a primitive $(2n+1)$-th root of unity, is a Frobenius-stable base with an algebraic membership test. Falsifier: at $n = 23$, "fail if the cyclotomic system's solving degree is not lower, or its yield per column is lower". The encyclopedia recorded the falsifier as not established; the 2026-10-12 census sampled 32 targets with no null population and its $n = 23$ solver cells timed out.

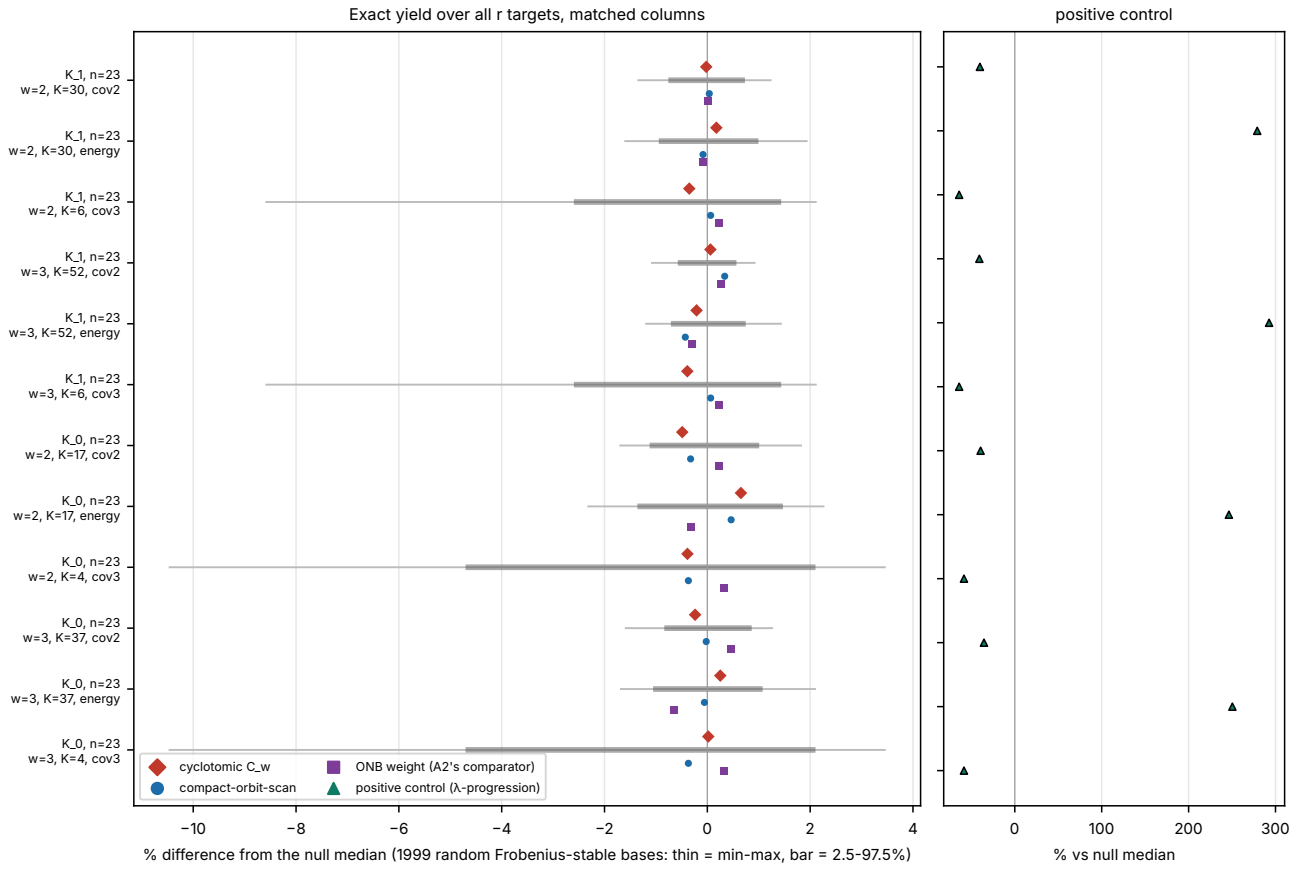
2. Method

Every base is moved to discrete-log space (full table of $[k]G$) and measured exactly over all r targets: $\mathrm{cov}_2 = |B + B|$, $\mathrm{cov}_3 = |B + B + B|$, and the additive energy. The mean number of ordered m-decompositions per target is $|B|^m/r$ for every base (identity), so at matched columns yield can differ only through sum collisions. Cyclotomic and compact bases are the `ic_framework` plug-ins' own output. Null: 1999 bases from uniform abscissae with identical lifting, cofactor projection and Frobenius/negation fold. Positive control: the λ-progression base. Cells at the transition size $K_m = \mathrm{round}\big((m!r\ln 2)^{1/m}/2n\big)$ capped at the base size. One-column bases are all unit multiples of $\{\pm\lambda^j\}$ and hence identical (amendment A1): C_1 cannot differ from any 1-column base.

3. Yield (measured, exact; N = 1999; critical p = 0.05/15)

rung	w	K	metric	C_w effect	p	null 2.5–97.5%	pos. control
K1 n23	2	30	cov2	−0.022%	0.951	−0.76, +0.73%	−40.1%
	2	30	energy	+0.176%	0.704	−0.94, +1.00%	+279%
	2	6	cov3	−0.351%	0.728	−2.60, +1.44%	−64.0%
	3	52	cov2	+0.060%	0.842	−0.58, +0.57%	−40.8%
	3	52	energy	−0.210%	0.553	−0.71, +0.75%	+293%
	3	6	cov3	−0.389%	0.700	−2.60, +1.44%	−64.0%
K0 n23	2	17	cov2	−0.488%	0.392	−1.12, +1.01%	−39.3%
	2	17	energy	+0.653%	0.363	−1.36, +1.47%	+246%
	2	4	cov3	−0.387%	0.828	−4.70, +2.10%	−58.5%
	3	37	cov2	−0.238%	0.582	−0.84, +0.86%	−35.5%
	3	37	energy	+0.252%	0.644	−1.05, +1.08%	+250%
	3	4	cov3	+0.018%	0.996	−4.70, +2.10%	−58.5%

$w = 1$: one column at K_1 (identical to every 1-column base, p = 1); no subgroup point at K_0 (no base). Compact-orbit-scan and the ONB comparator are also inside the null on every cell (README §3). At $n = 11$ ($r = 991$) the positive control is not detected at $K = 2$, so those cells are uninformative.



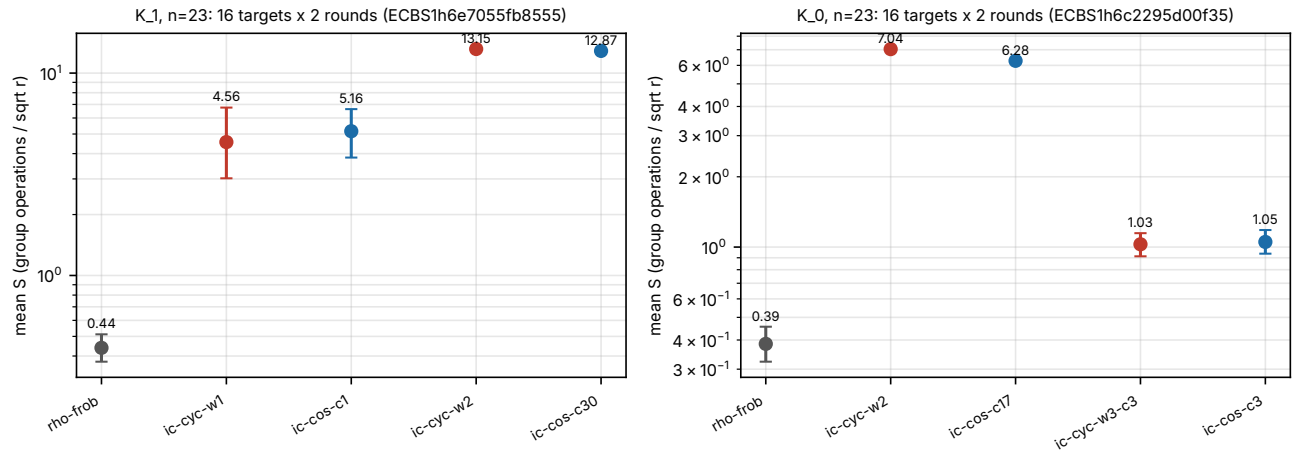
Black edge = significant at p < 0.05/15 (two-sided rank). Source: yield/yield-*.json (examples/cyclotomic_falsifier_yield.rs, seed 20261015). K=1 cells omitted: identical for every base; n=11 in the README table.

Figure 1: Exact yield relative to the null median, $n = 23$ cells; thin line = null min–max, bar = 2.5–97.5%; black edge = significant. Source `yield/yield-*.json`.

4. Cost (measured, ecbench, operations only, L0; IC S are lower bounds)

curve	cyclotomic / compact, equal columns	ratio of S [95%]	verdict
K1 n23	`w=1 / columns=1`	0.884 [0.544, 1.413]	unresolved
K1 n23	`w=2 (30) / columns=30`	1.022 [1.021, 1.023]	cyclotomic costlier
K0 n23	`w=2 (17) / columns=17`	1.122 [1.120, 1.125]	cyclotomic costlier
K0 n23	`w=3,columns=3 / columns=3`	0.977 [0.818, 1.161]	unresolved

Sessions `ECBS1h6e7055fb8555`, `ECBS1h6c2295d00f35`: 16 public targets × 2 rounds, 240/240 verified, replay-all 160/160 identical each. The costlier pairs differ in the factor-base phase (2247 vs 1661 and 2186 vs 1079 GAE in one record each) with identical oracle setup — build engineering, not yield (derived). Every IC arm is 2.7–30× rho-frob.



ecbench table, operations only (ic.pipeline, rho.signed_frobenius), L0; bars = recorded 95% interval. Red: cyclotomic; blue: compact-orbit-scan at equal columns.

Figure 2: Mean S per arm with recorded 95% intervals.

5. Limits

The solver clause was not re-run; its only record (`a2_summary_n11.md`: F4 max degree 4 vs 3, 754.6 vs 7.3 ms, Macaulay 0/16 vs 3/16 resolved at $n = 11$, $w = 1$, $m = 2$) goes against the lead but is not at A2′s $n = 23$. $n = 83$: no honest cheap cell (single-word plug-in; $r \approx 2^{81}$). Metrics cover $m \leq 3$ in the projected-subgroup model at transition sizes; not raw bases, $m \geq 4$, or other sizes. No security or asymptotic claim. Run report: `reports/run-20261015T0728Z/report.pdf`.